How to generate all possible rational Wilf–Zeilberger forms?

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Shift and difference operators

K: an algebraically closed field of characteristic 0.

$K(\mathbf{x})$: the field of rational functions in $\mathbf{x} = \{x_1, \ldots, x_n\}$ over K.

Shift operator:

$$\sigma_i(f(x_1, \ldots, x_n)) = f(x_1, \ldots, x_i + 1, \ldots, x_n),\ 1 \leq i \leq n.$$

Difference operator:

$$\Delta_i(f(\mathbf{x})) = \sigma_i(f(\mathbf{x})) - f(\mathbf{x}),\ 1 \leq i \leq n.$$

Hypergeometric terms

Definition

A nonzero term $H : \mathbb{N}^n \to K$ is hypergeometric over $K(\mathbf{x})$ if $\exists f_1, \ldots, f_n \in K(\mathbf{x})$ s.t.

$$\frac{\sigma_i(H)}{H} = f_i$$

for all $x \in \mathbb{N}^n$ and $1 \leq i \leq n$. Call $f_1, \ldots, f_n$ the certificates of H.

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Two hypergeometric terms H_1, H_2 are conjugate if they have the same certificates.

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Compatibility conditions:

$$\frac{\sigma_i(f_j)}{f_j} = \frac{\sigma_j(f_i)}{f_i}, \ 1 \leq i, j \leq n.$$

Hyperarithmetic terms

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A nonzero term $H : \mathbb{N}^n \to K$ is hyperarithmetic over $K(\mathbf{x})$ if $\exists f_1, \ldots, f_n \in K(\mathbf{x})$ s.t.

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Additive compatibility conditions:

$$\sigma_i(f_j) - f_j = \sigma_j(f_i) - f_i, \ 1 \leq i, j \leq n.$$

We call such an n tuple $(f_1, \ldots, f_n)$ a "rational" WZ-form.

Ore-Sato theorem

$$\prod_s^t \alpha_i = \begin{cases} \prod_{i=s}^{t-1} \alpha_i, & \text{if } t \geq s; \\ \prod_{i=t}^{s-1} \alpha_i^{-1}, & \text{if } t < s. \end{cases}$$

Theorem

If $f_1, \ldots, f_n \in K(\mathbf{x})$ *satisfy the following compatibility conditions*

$$\sigma_i(f_j)/f_j = \sigma_j(f_i)/f_i, \quad 1 \leq i, j \leq n,$$

then

$$f_i = \frac{\sigma_i(a)}{a} u_i \prod_{\mathbf{v} \in V} \prod_{\ell}^{v_j} r_{\mathbf{v}}(\mathbf{v} \cdot \mathbf{x} + \ell),$$

where $a \in K(\mathbf{x})$, $u_1, \ldots, u_n \in K$, $V \subset \mathbb{Z}^n$ *with* $|V| < +\infty$, *and for each* $\mathbf{v} = (v_1, \ldots, v_n) \in V$, $r_{\mathbf{v}} \in K(z)$.

Rising factorials and factorial terms

$$(\alpha)_n = \begin{cases} \prod_{i=0}^{n-1}(\alpha+i), & n \geq 0, \\ \prod_{i=1}^{-n}\frac{1}{\alpha-i}, & n<0 \text{ and } \alpha \neq 1,2,\ldots,-n, \\ 0, & \textit{otherwise.} \end{cases}$$

Definition

A nonzero term $T : \mathbb{N}^n \to K$ is a factorial term if there are $\mathbf{u} \in K^n$, $p, q \in \mathbb{N}$, $\alpha \in K^{p+q}$, and $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p \in \mathbb{Z}^n$ s.t.

$$T(\mathbf{x}) = \mathbf{u}^{\mathbf{x}} \prod_{i=1}^{p} (\alpha_i)_{\mathbf{v}_i \cdot \mathbf{x}} \prod_{i=p+1}^{p+q} (\alpha_i + \mathbf{v}_i \cdot \mathbf{x})_{-\mathbf{v}_i \cdot \mathbf{x}}$$

for all $\mathbf{x} \in \mathbb{N}^n$, where $\mathbf{u}^{\mathbf{x}} := u_1^{x_1} \cdots u_n^{x_n}$, $\mathbf{v}_i \cdot \mathbf{x} = v_{i,1}x_1 + \ldots + v_{i,n}x_n$.

Ore-Sato theorem

$$f_i = \frac{\sigma_i(a)}{a} \cdot u_i \prod_{\mathbf{v}\in V} \prod_{\ell}^{v_j}{}_0 \, r_{\mathbf{v}}(\mathbf{v}\cdot\mathbf{x}+\ell),$$

Corollary (Structure of multivariate hypergeometric terms)

Any hypergeometric term H is conjugate to

$$R(\mathbf{x}) \cdot T(\mathbf{x}),$$

where $R \in K(\mathbf{x}) \setminus \{0\}$ and T is a nontrivial factorial term.

Goal. Extending this theorem to the additive case, *i.e.*, finding the structure of $f_i's$ if $\Delta_i(f_j) = \Delta_j(f_i)$.

Cyclic operators

$G := \langle \sigma_1, \ldots, \sigma_n \rangle$: free abelian group generated by $\{\sigma_1, \ldots, \sigma_n\}$.

Cyclic operator:

For $n \in \mathbb{Z}$ and $\theta \in \langle \sigma_1, \ldots, \sigma_n \rangle$, define

$$\frac{\theta^n - 1}{\theta - 1} = \begin{cases} \sum_{j=0}^{n-1} \theta^j, & n \geq 0; \\ -\sum_{j=1}^{-n} \theta^{-j}, & n < 0. \end{cases}$$

Note that $\frac{\theta^0 - 1}{\theta - 1} \bullet f = 0$ for any $f \in K(\mathbf{x})$.

Additive Ore-Sato theorem: the bivariate case

Theorem (Chen2019)

If $f, g \in K(x, y)$ *satisfy* $\Delta_y(f) = \Delta_x(g)$, *then*

$$f = \Delta_x(a) + \sum_{i=1}^{m} \frac{\sigma_y^{s_i} - 1}{\sigma_y - 1} \bullet b_i,$$

$$g = \Delta_y(a) + \sum_{i=1}^{m} \frac{\sigma_x^{t_i} - 1}{\sigma_x - 1} \bullet b_i,$$

where $a, b_1, \ldots, b_m \in K(x, y)$ *with for each* i, $\sigma_y^{s_i}(b_i) = \sigma_x^{t_i}(b_i)$ *for some* $s_i \in \mathbb{Z}_{\geq 0}$, $t_i \in \mathbb{Z}$ *with* s_i, t_i *not all zero.*

Additive Ore-Sato theorem: the multivariate case

$$\sum_i^t \alpha_i = \begin{cases} \sum_{i=s}^{t-1} \alpha_i, & \text{if } t \geq s; \\ -\sum_{i=t}^{s-1} \alpha_i, & \text{if } t < s. \end{cases}$$

Theorem

If $f_1, \ldots, f_n \in K(\mathbf{x})$ *satisfy the additive compatibility conditions*

$$\sigma_i(f_j) - f_j = \sigma_j(f_i) - f_i, \ 1 \leq i, j \leq n,$$

then

$$f_i = \Delta_i(a) + \mu_i + \sum_{\mathbf{v} \in V} \sum_{\ell}^{v_i} r_{\mathbf{v}}(\mathbf{v} \cdot \mathbf{x} + \ell), \ \text{for } i = 1, \ldots, n,$$

where $a \in K(\mathbf{x})$, $\mu_1, \ldots, \mu_n \in K$, $V \subset \mathbb{Z}^n$ *with* $|V| < +\infty$, *and for each* $\mathbf{v} = (v_1, \ldots, v_n) \in V$, $r_{\mathbf{v}} \in K(z)$.

Sketch of the proof:

Theorem (Structure of WZ-forms)

Any WZ-form can be decomposed into one exact WZ-form plus several uniform WZ-forms.

Exact WZ-form: A WZ-form $(f_1, \ldots, f_n)$ with respect to $(\Delta_1, \ldots, \Delta_n)$ is said to be *exact* if there exists $g \in K(\mathbf{x})$ such that $f_i = \Delta_i(g)$, for all $i \in \{1, \ldots, n\}$.

Uniform WZ-form: A WZ-form $(f_1, \ldots, f_n)$ with respect to $(\Delta_1, \ldots, \Delta_n)$ is called a *uniform WZ-form* if there exists an integer vector $\mathbf{v}$ and for each i a univariate function $r_i \in K(z)$ s.t. $f_i = r_i(\mathbf{v} \cdot \mathbf{x})$.

Sketch of the proof:

Notation:

$$G := \langle \sigma_1, \ldots, \sigma_n \rangle, \quad \widehat{\mathbf{x}} := x_2, \ldots, x_n.$$

Orbital decomposition w.r.t. x_1 and G:

$$f_1 = p + \sum_{i=1}^{I} \sum_{j=1}^{J} \sum_{\tau \in \Lambda_{i,j}} \frac{a_{i,j,\tau}}{\tau(b_i^j)},$$

where $p, a_{i,j,\tau}, b_i \in K(\widehat{\mathbf{x}})[x_1]$ with $\Lambda_{i,j} \subset G$, $b_i \nsim_G b_{i'}$, and we have $\deg_{x_1}(a_{i,j,\tau}) < \deg_{x_1}(b_i)$.

Sketch of the proof:

Fact.

$$\frac{a}{\phi^m(u)} = \phi(g) - g + \frac{\phi^{-m}(a)}{u},$$

Abramov's reduction:

$$f_1 = \Delta_1(g_0) + \sum_{i=1}^{I}\sum_{j=1}^{J}\widetilde{f}_{1,i,j} \quad \text{with} \quad \widetilde{f}_{1,i,j} = \sum_{\lambda\in\widetilde{\Lambda}_{i,j}} \frac{\widetilde{a}_{i,j,\lambda}}{\lambda(b_i^j)},$$

where $g_0 \in K(\mathbf{x})$, $\widetilde{\Lambda}_{i,j} \subseteq \langle \sigma_2, \ldots, \sigma_n \rangle$, and $\lambda(b_i) \not\sim_{\langle\sigma_1\rangle} \lambda'(b_i)$.

Sketch of the proof:

Theorem

If $\Delta_k(f_1)$ is σ_1-summable for each $k \in \{2, \ldots, n\}$, then for each i, j, $\Delta_k(\widetilde{f}_{1,i,j})$ is σ_1-summable, and for each i, all of the $\widetilde{a}_{i,j,\lambda}$ and b_i are integer-linear of the same type.

Definition (Integer-linearity)

$f \in K(\mathbf{x})$ is called integer-linear of type $\mathbf{v}$ if it can be written as

$$f = r(\mathbf{v} \cdot \mathbf{x}),$$

where $\mathbf{v} \in \mathbb{Z}^n$ and $r \in K(z)$ is a univariate rational function.

Sketch of the proof:

Now we focus on each i, j, i.e., let $f_1 = \sum_{\lambda\in\Lambda} a_\lambda/\lambda(b^m)$, where $\Lambda \subset \langle\sigma_2, \ldots, \sigma_n\rangle$ and $\lambda(b) \not\sim_{\langle\sigma_1\rangle} \lambda'(b)$.
For each $k \in \{2, \ldots, n\}$, $\Lambda_k \subset \langle\sigma_2, \ldots, \hat{\sigma}_k, \ldots, \sigma_n\rangle$,

$$f_1 = \Delta_1(f_{1,k}) + \frac{\sigma_k^{L_k} - 1}{\sigma_k - 1} \bullet \sum_{\eta\in\Lambda_k} \frac{\widetilde{a}_{\eta,0}}{\eta(b^m)}.$$

Let

$$f_k' = \Delta_k(f_{1,k}) + \frac{\sigma_1^{N_k} - 1}{\sigma_1 - 1} \bullet \sum_{\eta\in\Lambda_k} \frac{\widetilde{a}_{\eta,0}}{\eta(b^m)},$$

where $\sigma_k^{L_k}(\widetilde{a}_{\eta,0}) = \sigma_1^{N_k}(\widetilde{a}_{\eta,0})$ and $\sigma_k^{L_k}(b) = \sigma_1^{N_k}(b)$, then we can build a uniform WZ-form $(f_1, f_2', \ldots, f_n')$, then induction follows.

Sketch of the proof:

Theorem (Structure of uniform WZ-forms)

Let $\big(f_1(\mathbf{v}\cdot\mathbf{x}),\dots,f_n(\mathbf{v}\cdot\mathbf{x})\big)$ *be a uniform WZ-form of the type* $\mathbf{v}$, *then there exist constants* $\mu_1,\dots,\mu_n\in K$ *and a univariate rational function* $r\in K(z)$ *such that for each* $i\in\{1,\dots,n\}$,

$$f_i(\mathbf{v}\cdot\mathbf{x})=\mu_i+\sum_{\ell}^{v_i}{}_0\, r(\mathbf{v}\cdot\mathbf{x}+\ell).$$

Structure of WZ-forms:

$$f_i=\Delta_i(a)+\mu_i+\sum_{\mathbf{v}\in V}\sum_{\ell}^{v_i}{}_0\, r_{\mathbf{v}}(\mathbf{v}\cdot\mathbf{x}+\ell),\ \text{for } i=1,\dots,n,$$

Corollary:

$$r(z) = \sum_s \sum_t \frac{\beta_{s,t}}{(z+\alpha_s)^t},$$

Polygamma function:

$$\psi^{(t)}(z+\alpha_s+1) - \psi^{(t)}(z+\alpha_s) = \frac{(-1)^t t!}{(z+\alpha_s)^{t+1}}.$$

Theorem (Structure of multivariate hyperarithmetic terms)

Any hyperarithmetic term is conjugate to

$$a + \sum_{\mathbf{v}\in V} \sum_s \sum_t \beta_{\mathbf{v},s,t} \psi^{(t)}(\mathbf{v}\cdot\mathbf{x} + \alpha_{\mathbf{v},s}),$$

where $a \in K(\mathbf{x})$, $V \subset \mathbb{Z}^n$, $s, t \in \mathbb{N}$, *and for each* $\mathbf{v}$, $\beta_{\mathbf{v},s,t}, \alpha_{\mathbf{v},s} \in K$.

Example:

Let $\omega = (f, g, h) \in K(x, y, z)^3$ be a WZ-form with

$$f = \sum_{\ell=0}^{3} \frac{1}{4x + 6y + 5z + \ell},$$

$$g = \sum_{\ell=0}^{5} \frac{1}{4x + 6y + 5z + \ell} + \sum_{\ell=0}^{2} \frac{1}{3y + 2z + \ell},$$

$$h = \sum_{\ell=0}^{4} \frac{1}{4x + 6y + 5z + \ell} + \sum_{\ell=0}^{1} \frac{1}{3y + 2z + \ell}.$$

Then the corresponding hyperarithmetic term H is conjugate to

$$\psi^{(0)}(4x + 6y + 5z) + \psi^{(0)}(3y + 2z).$$

Some related results:

$$\partial_i := \frac{\partial}{\partial_{x_i}}, \text{ for } i = 1, \ldots, n.$$

Theorem (Multivariate Christopher's theorem)

Let $f_1, \ldots, f_n \in K(\mathbf{x})$ be rational functions such that $\partial_i(f_j) = \partial_j(f_i)$ for $1 \leq i < j \leq n$. Then there exist rational functions $a \in K(\mathbf{x})$, $g_1, \ldots, g_L \in K(\mathbf{x})$ and constants $\gamma_1, \ldots, \gamma_L \in K$ such that

$$f_i = \partial_i(a) + \sum_{\ell=1}^{L} \gamma_\ell \frac{\partial_i(g_\ell)}{g_\ell}, \text{ for } i = 1, \ldots, n.$$

Any multivariate hyperexponential term is conjugate to $exp(a) \cdot \prod_{\ell=1}^{L} g_\ell^{\gamma_\ell}$.

Some related results:

$$\tau_i(q_i^{y_i}) = q_i^{y_i+1}, \text{ for } i = 1, \ldots, n. \quad F := K(q_1^{y_1}, \ldots, q_n^{y_n}).$$

Theorem (Ore–Sato theorem in the q-difference case)

Let H be a q-hypergeometric term over F. Then there exist a rational function $a \in F$ and a q-factorial term G such that $H = a \cdot G$.

Definition (q-integer-linearity)

Let $P = \{P_1, \ldots, P_k\}$ be the partition of $q_1, \ldots, q_n$. An irreducible polynomial $h \in K[q_1^{y_1}, \ldots, q_n^{y_n}]$ is said to be q-integer-linear if there exist an integer $t \in \{1, 2, \ldots, k\}$, two integer-linear polynomials $\sum_{s \in P_t} i_s y_s$ and $\sum_{s \in P_t} j_s y_s$ and a univariate polynomial $P \in K[z]$ such that

$$h = q^{\sum_{s \in P_t} i_s y_s} P(q^{\sum_{s \in P_t} j_s y_s}), \ q \text{ is the signature of } P_t.$$

Thank you!